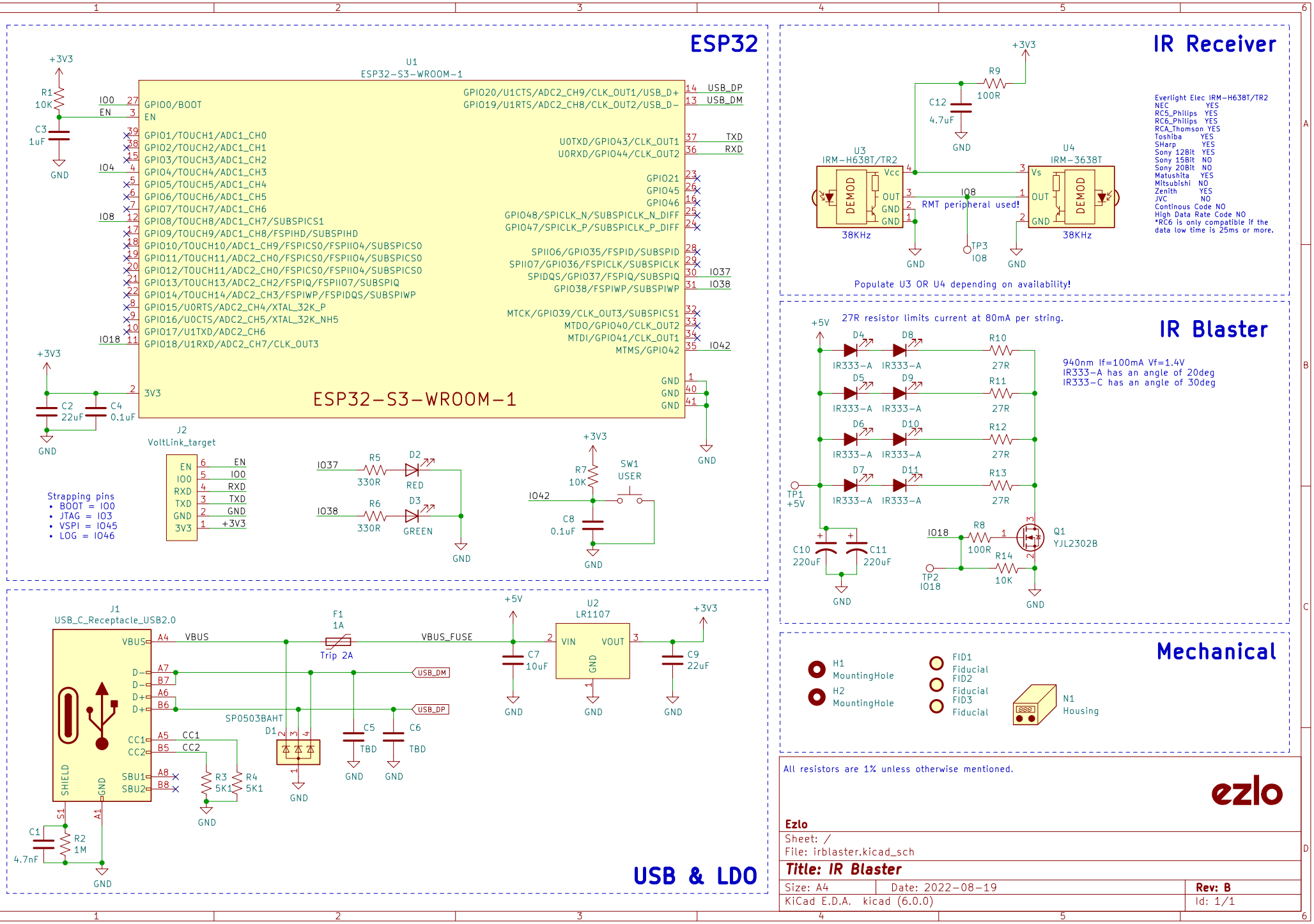

ESP32
U1
ESP32-S3-WROOM-1
ESP32-S3-WROOM-1
+3V3
R1
10K
C3
1uF
GND
IO0
EN
IO4
IO8
IO18
C2
22uF
C4
0.1uF
3V3
GPIO0/BOOT
EN
GPIO1/TOUCH1/ADC1_CH0
GPIO2/TOUCH2/ADC1_CH1
GPIO3/TOUCH3/ADC1_CH2
GPIO4/TOUCH4/ADC1_CH3
GPIO5/TOUCH5/ADC1_CH4
GPIO6/TOUCH6/ADC1_CH5
GPIO7/TOUCH7/ADC1_CH6
GPIO8/TOUCH8/ADC1_CH7/SUBSPICS1
GPIO9/TOUCH9/ADC1_CH8/FSPIHD/SUBSPIHD
GPIO10/TOUCH10/ADC1_CH9/FSPICS0/FSPIIO4/SUBSPICS0
GPIO11/TOUCH11/ADC2_CH0/FSPICS0/FSPIIO4/SUBSPICS0
GPIO12/TOUCH11/ADC2_CH0/FSPICS0/FSPIIO4/SUBSPICS0
GPIO13/TOUCH13/ADC2_CH2/FSPIQ/FSPIIO7/SUBSPIQ
GPIO14/TOUCH14/ADC2_CH3/FSPIWP/FSPIDQS/SUBSPIWP
GPIO15/U0RTS/ADC2_CH4/XTAL_32K_P
GPIO16/U0CTS/ADC2_CH5/XTAL_32K_NH5
GPIO17/U1TXD/ADC2_CH6
GPIO18/U1RXD/ADC2_CH7/CLK_OUT3
GPIO20/U1CTS/ADC2_CH9/CLK_OUT1/USB_D+
GPIO19/U1RTS/ADC2_CH8/CLK_OUT2/USB_D-
U0TXD/GPIO43/CLK_OUT1
U0RXD/GPIO44/CLK_OUT2
GPIO21
GPIO45
GPIO46
GPIO48/SPICLK_N/SUBSPICLK_N_DIFF
GPIO47/SPICLK_P/SUBSPICLK_P_DIFF
SPIIO6/GPIO35/FSPID/SUBSPID
SPIIO7/GPIO36/FSPICLK/SUBSPICLK
SPIDQS/GPIO37/FSPIQ/SUBSPIQ
GPIO38/FSPIWP/SUBSPIWP
MTCK/GPIO39/CLK_OUT3/SUBSPICS1
MTDO/GPIO40/CLK_OUT2
MTDI/GPIO41/CLK_OUT1
MTMS/GPIO42
USB_DP
USB_DM
TXD
RXD
IO37
IO38
IO42
GND
J2
VoltLink_target
EN
IO0
RXD
TXD
GND
3V3
Strapping pins
• BOOT = IO0
• JTAG = IO3
• VSPI = IO45
• LOG = IO46
IO37
R5
330R
D2
RED
IO38
R6
330R
D3
GREEN
+3V3
R7
10K
SW1
USER
IO42
C8
0.1uF
GND
IR Receiver
+3V3
R9
100R
C12
4.7uF
GND
U3
IRM-H638T/TR2
DEMOD
Vcc
OUT
GND
38KHz
U4
IRM-3638T
Vs
OUT
GND
38KHz
IO8
RMT peripheral used!
TP3
IO8
Everlight Elec IRM-H638T/TR2
NEC YES
RC5_Philips YES
RC6_Philips YES
RCA_Thomson YES
Toshiba YES
SHarp YES
Sony 12Bit YES
Sony 15Bit NO
Sony 20Bit NO
Matushita YES
Mitsubishi NO
Zenith YES
JVC NO
Continous Code NO
High Data Rate Code NO
*RC6 is only compatible if the data low time is 25ms or more.
Populate U3 OR U4 depending on availability!
IR Blaster
+5V
27R resistor limits current at 80mA per string.
D4
D5
D6
D7
D8
D9
D10
D11
IR333-A
R10
R11
R12
R13
27R
940nm If=100mA Vf=1.4V
IR333-A has an angle of 20deg
IR333-C has an angle of 30deg
TP1
+5V
C10
220uF
C11
220uF
GND
IO18
R8
100R
Q1
YJL2302B
R14
10K
TP2
IO18
Mechanical
H1
MountingHole
H2
MountingHole
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
N1
Housing
USB & LDO
J1
USB_C_Receptacle_USB2.0
VBUS
D-
D+
CC1
CC2
SBU1
SBU2
SHIELD
GND
A4
A7
B7
A6
B6
A5
CC1
B5
CC2
A8
B8
S1
A1
F1
1A
Trip 2A
VBUS_FUSE
USB_DM
USB_DP
SP0503BAHT
D1
C5
TBD
C6
TBD
R3
5K1
R4
5K1
C1
4.7nF
R2
1M
+5V
U2
LR1107
VIN
VOUT
GND
C7
10uF
C9
22uF
+3V3
All resistors are 1% unless otherwise mentioned.
ezlo
Ezlo
Sheet: /
File: irblaster.kicad_sch
Title: IR Blaster
Size: A4
Date: 2022-08-19
Rev: B
KiCad E.D.A. kicad (6.0.0)
Id: 1/1